

实验作业2

一、实验背景与目标

二、数据集与特征

2.1 数据来源

2.2 特征

2.2.1 项目

2.2.2 PR作者

2.2.3 PR评审者

2.2.4 代码变更

三、建模方法与评价指标

3.1 任务一：预测 Pull Request 关闭时长

3.2 任务二：预测 Pull Request 合入与否

四、难度梯度

六、提交清单

参考文献

以小组为单位提交，在报告首页写清每位同学的学号、姓名、在作业中的具体工作内容与贡献

数据集下载链接：<https://pan.quark.cn/s/e3b53180bd3c>

实验作业1中使用了传统机器学习方法完成了 Pull Request (PR) 的关闭时长与合入预测。本实验在此基础上，要求必须使用神经网络方法完成建模与预测。

一、实验背景与目标

- 任务一：预测 Pull Request 关闭时长
 - 定义：从 PR创建到被合入或关闭的间隔。
 - 意义：合理预估处理时长可以帮助维护者优先处理紧急PR，提高代码评审效率。
- 任务二：预测 Pull Request 合入与否
 - 定义：对每个PR进行二分类，判断其是被合并（merged）还是在评审后关闭但未合并（closed）。

- **意义**：帮助维护者提前识别可能被拒绝的PR，从而改进贡献者的提交或分配评审资源。

二、数据集与特征

2.1 数据来源

课程提供 10 个开源项目数据（实验1已有数据 + 新增9个项目），均为 PR 级别的统计信息，并包含已计算完成的特征。为保证实验的公平性和防止数据泄漏，推荐按时间顺序分割训练集和测试集。例如将训练集设为**项目开始**至 2021 年 5 月 31 日、测试集设为 2021 年 6 月 1 日至 2022 年 6 月 15 日。在进行实验时，应使用类似的时间切分策略，而不是随机划分样本。

2.2 特征

⚠ ⚠ **重要提醒** ⚠ ⚠ :

实验所提供的数据集与论文中的特征并非一一对应，存在字段缺失、命名不一致或统计口径差异。需要同学们**自行对照论文定义与数据集字段**，完成合理的特征映射。需要对异常值、缺失值进行清理，不能直接拿到特征表就当作输入。允许同学根据理解对部分特征进行组合、转换或舍弃，但需在实验报告中说明理由。注：在GitHub, 合入请求是pull request; 在GitLab, 合入请求是merge request; 在Gerrit, 合入请求是change, 三者含义相同。

2.2.1 项目

Feature	Description
General Information	
project_age	The number of days since the first change was submitted.
language_num	The number of programming languages used in this project.
change_num	The number of changes in the project.
open_changes	The number of open changes in the project.
author_num	The number of authors in the project.
reviewer_num	The number of human reviewers in the project.
team_size	The number of participants (authors, reviewers) in the project.
changes_per_author	The number of changes submitted per author in the project.

changes_per_reviewer	The number of changes reviewed per reviewer in the project.
changes_per_week	The number of changes submitted per week in the project.
Code Changes	
avg_lines	The average lines of code in a change in the project.
avg_segs	The average segments in a change in the project.
avg_files	The average files in a change in the project.
add_per_week	The lines of code added per week in the project.
del_per_week	The lines of code deleted per week in the project.
Review Experience	
merge_proportion	The proportion of the merged changes of the project.
avg_reviewers	The average reviewers assigned to a change in the project.
avg_comments	The average review comments of a change in the project.
avg_iterations	The average review iterations of a change in the project.
avg_duration	The average review duration of a change in the project.
avg_score	The average review scores of a change in the project.
avg_iterations_merged	The average review iterations of the merged changes in the project.
avg_duration_merged	The average review duration of the merged changes in the project.
avg_churn_merged	The average churn of the merged changes in the project.
avg_files_merged	The average files of the merged changes in the project.
avg_comments_merged	The average review comments of the merged changes in the project.
avg_iterations_abandoned	The average iterations per abandoned changes in the project.
avg_duration_abandoned	The average duration per abandoned changes in the project.
avg_churn_abandoned	The average churn per abandoned changes in the project.

avg_files_abandoned	The average files per abandoned changes in the project.
avg_comments_abandoned	The average review comments per abandoned changes in the project.

2.2.2 PR作者

Feature	Description
General Information	
experience	The years since the author submitted the first change.
is_reviewer	Is the author also a reviewer?
Review Experience	
change_num	The number of changes submitted by the author.
participation	The proportion of the changes in the project (all projects) submitted by the author.
changes_per_week	The number of changes submitted per week by the author.
avg_iterations	The average review iterations of the changes submitted by the author.
avg_duration	The average review duration of the changes submitted by the author.
avg_score	The average review score of the changes submitted by the author.
merge_proportion	The proportion of the merged changes submitted by the author.
Collaboration Network	
degree centrality	The degree centrality of the author in the collaboration network.
closeness centrality	The closeness centrality of the author in the collaboration network.
betweenness centrality	The betweenness centrality of the author in the collaboration network.
eigenvector centrality	The eigenvector centrality of the author in the collaboration network.

clustering_coefficient	The clustering centrality of the author in the collaboration network.
k_coreness	The k-coreness of the author in the collaboration network.

2.2.3 PR评审者

Feature	Description
General Information	
experience	The year since the reviewer submitted his first review comments.
is_author	Is the reviewer also an author?
Review Experience	
change_num	The number of changes reviewed by the reviewer.
participation	The proportion of the changes in the project (all projects) reviewed by the reviewer.
avg_comments	The average review comments left in a change the reviewer involved.
avg_files	The average reviewed files in a change the reviewer involved.
avg_iterations	The average review iterations of a change the reviewer involved.
avg_duration	The average review duration of a change the reviewer involved.
avg_score	The average review score of a change the reviewer involved.
merge_proportion	The proportion of the merged changes the reviewer involved.
Collaboration Network	
degree_centrality	The degree centrality of the reviewer in the collaboration network.
closeness_centrality	The closeness centrality of the reviewer in the collaboration network.
betweenness_centrality	The betweenness centrality of the reviewer in the collaboration network.
eigenvector_centrality	The eigenvector centrality of the reviewer in the collaboration network.

clustering_coefficient	The clustering centrality of the reviewer in the collaboration network.
k_coreness	The k-coreness of the reviewer in the collaboration network.

2.2.4 代码变更

Feature	Description
General Information	
directory_num	The number of directories in the change.
subsystem_num	The number of subsystems in the change.
language_num	The number of programming languages in the change.
file_type	The number of file types in the change.
has_test	Whether the change includes test files?
Textual Information Subject \& Messages	
has_feature	Whether the word "feature" occurs in the subject of the change?
has_bug	Whether the word "bug" occurs in the subject of the change?
has_document	Whether the word "document" occurs in the subject of the change?
has_improve	Whether the word "improve" occurs in the subject of the change?
has_refactor	Whether the word "refactor" occurs in the subject of the change?
subject_length	The number of words in the subject of the change.
subject_readability	The Coleman–Liau readability of the subject of the change.
subject_embedding	The BERT embedding of the subject of the change.
message_length	The number of words in the message of the change.
message_readability	The Coleman–Liau readability of the message of the change.
message_embedding	The BERT embedding of the message of the change.
Code Changes	
lines_added	The lines of code added in the change.

lines_deleted	The lines of code deleted in the change.
segs_added	The number of code segments added in the change.
segs_deleted	The number of code segments deleted in the change.
segs_updated	The number of code segments updated in the change.
files_added	The number of files added in the change.
files_deleted	The number of files deleted in the change.
files_updated	The number of files updated in the change.
modify_proportion	The proportion of modified lines in the change.
modify_entropy	The distribution of modified code across files in the change.
test_churn	The lines of code modified in the test files in the change.
non_test_churn	The lines of code modified in the non-test files in the change.
Review Experience	
reviewer_num	The number of reviewers assigned to the change.
bot_reviewer_num	The number of bot reviewers assigned to the change.
has_reviewed	Has this change been reviewed?
comment_num	The number of review comments of the change.
comment_length	The number of words of the review comments of the change.
comment_embedding	The BERT embedding of the review comments of the change.
last_comment_mention	Whether the last review comment of the change contains a user mention (i.e. @)?

三、建模方法与评价指标

3.1 任务一：预测 Pull Request 关闭时长

- 建模方法：神经网络模型，如：Wide&Deep、DeepCross、Shared-Bottom、MMoE

- 评价指标：平均绝对误差（MAE）、均方误差（MSE）、均方根误差（RMSE）、决定系数（ R^2 ）等

3.2 任务二：预测 Pull Request 合入与否

- 建模方法：神经网络模型，如：Wide&Deep、DeepCross、Shared-Bottom、MMoE
- 评价指标：准确率(Accuracy)、精确度(Precision)、召回率(Recall)、F1分数（Macro平均）等

四、难度梯度

为了确保每位同学都能从实验中受益，我们将任务分为三个难度层级：

1. 基础层（Level 1）：使用给定的项目数据与部分特征完成神经网络模型，并报告实验过程与结果。（70 分）
2. 进阶层（Level 2）：（20 分）
 - a. 特征工程（例如：归一化）
 - b. 消融实验（例如：输入特征消融、模型组件消融等）
 - c. 泛化性实验（例如：跨项目预测）
 - d. 使用假设检验评估模型
3. 拓展层（Level 3）：实现一个多任务学习(multi-task learning)神经网络，同时预测两个任务。（10 分）

达到更高级别可以获得额外分数，但完成基础层（Level 1）是实验合格的基本条件。

六、提交清单

提交材料应包括：

1. 完整的实验报告建议包含以下部分，可以适当调整结构：
 - a. 任务一：预测 Pull Request 处理时间
 - Level 1
 - i. 问题与数据：明确任务定义，说明所用仓库数量、数据来源、时间切分方式、防止数据泄漏的措施。
 - ii. 特征：列出所有使用的特征。
 - iii. 模型与方法：描述使用的神经网络模型。
 - iv. 结果与分析：提供所有指标的可视化图表并进行结果分析。

v. **结论与建议**：总结发现，并提出对项目维护者或代码评审流程的建议。

▪ **Level 2**（在Level 1基础上需额外提交）

i. **特征**：特征工程说明。

ii. **模型与方法**：针对泛化性实验（例如：跨项目预测，需要说明训练集与测试集项目的选择与切分）、假设检验方法。

iii. **结果与分析**：提供消融实验结果、泛化性实验结果、假设检验结果。

▪ **Level 3**（在Level 1基础上需额外提交）

iv. **模型与方法**：多任务神经网络模型说明。

v. **结果与分析**：多任务实验结果，包括与单任务模型效果对比等。

b. 任务二：预测 Pull Request 结局

i. 要求与任务一相同（Level 1/2/3 结构一致）。

2. 源代码、原始数据、处理后的特征数据等，可用 README 指导助教复现实验

作业提交截止日期：10 月 31 日

参考文献

[1] Yang L, Zhang H, Xu J, et al. A preliminary investigation on using multi-task learning to predict change performance in code reviews. Empirical Software Engineering, 2024, 157(29): 41:1–41:53

[2] Wide&Deep: https://www.zhihu.com/tardis/zm/art/132708525?source_id=1005

[3] DeepCross: <https://zhuanlan.zhihu.com/p/368381633>

[4] Shared-Bottom、MMoE: <https://zhuanlan.zhihu.com/p/17814626067>

[5] 假设检验: <https://www.zhihu.com/question/27306416/answer/372241948>